

Smart Water Tank Pump Control

DDCA Project – Design and Simulation of a Smart Water Tank Overflow Protection and Pump Control System

Abstract

1. Smart Water Tank Pump Control is a digital system designed to automate water-level monitoring and pump operation.
2. The system helps prevent water tank overflow and unnecessary wastage of water.
3. It uses water-level sensors to detect the current level of water in the tank.
4. Based on the detected level, the control logic automatically turns the pump ON or OFF.
5. When the water level is low, the pump is switched ON to fill the tank.
6. When the tank reaches its maximum safe level, the pump is switched OFF automatically.
7. The system can be designed using Digital Design and Computer Architecture (DDCA) concepts such as logic gates, comparators, and flip-flops.
8. The proposed system reduces manual monitoring, power consumption, and water wastage.
9. The design can be implemented and verified through digital simulation before practical implementation.
10. Overall, the project provides a simple, reliable, and efficient solution for automated water tank overflow protection and pump control.